

Chapter 03: Higher-Order Linear Differential Equations & Mechanical Oscillators

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213)

Textbook:

Advanced Engineering Mathematics

(7th Edition) by Dennis G. Zill — Chapter 3 (§3.1, §3.2, §3.3, §3.4, §3.5, §3.6, §3.8)

1. Executive Overview & Structural Taxonomy

Higher-order linear differential equations govern physical systems with inertia, elasticity, energy dissipation, and external periodic forcing. While first-order equations describe relaxation, decay, and transient accumulation, second-order and higher-order equations capture **vibrations, waves, resonance, and multidimensional feedback loops**.

Higher-Order Linear Differential Equations (Chapter 3)

- 1. Theoretical Foundations (§3.1)
 - Existence & Uniqueness Theorem 3.1.1 (IVP vs BVP)
 - Superposition principle & linear independence
 - The Wronskian determinant $W(y_1, y_2, \dots, y_n)$ & Abel's formula
- 2. Solution of Homogeneous Equations
 - §3.2 Reduction of Order: Given y_1 , find y_2 via $y_2 = y_1 \int (e^{\int P dx})/y_1^2 dx$
 - §3.3 Constant Coefficients: $a y'' + b y' + c y = 0$ via $ar^2 + br + c = 0$
 - Case 1: Real distinct roots ($r_1 \neq r_2$)
 - Case 2: Repeated real root ($r_1 = r_2 \Rightarrow y = (c_1 + c_2 x)e^{rx}$)
 - Case 3: Complex conjugate roots ($\alpha \pm i\beta \Rightarrow e^{\alpha x}[c_1 \cos \beta x + c_2 \sin \beta x]$)
 - §3.6 Cauchy-Euler Equations: $a x^2 y'' + b x y' + c y = 0$ via $a m(m-1) + bm + c = 0$
- 3. Non-Homogeneous Equations ($y = y_c + y_p$)
 - §3.4 Method of Undetermined Coefficients (Superposition & Annihilator)
 - The Duplication Rule: Multiply trial terms by x^s when colliding with y_c
 - §3.5 Method of Variation of Parameters: $y_p = u_1 y_1 + u_2 y_2$
 - $u_1' = -y_2 f(x)/W$ and $u_2' = y_1 f(x)/W$
 - Handles non-elementary forcing ($\tan x, \sec x, 1/x, e^x/x$)
- 4. Engineering Applications (§3.8)
 - Mass-spring-damper mechanical systems: $m x'' + \beta x' + k x = F(t)$
 - Free vibrations: Underdamped, Critically Damped, Overdamped
 - Forced vibrations & Pure Resonance: Catastrophic linear amplitude growth ($t \cdot \sin \omega t$)

2. Theoretical Foundations of Linear Equations (§3.1)

2.1 Initial-Value vs. Boundary-Value Problems

An n -th order linear differential equation in normal standard form is written as:

$$y^{(n)} + P_{n-1}(x)y^{(n-1)} + \dots + P_1(x)y' + P_0(x)y = g(x)$$

A. Initial-Value Problem (IVP)

All conditions are prescribed at a **single point** x_0 :

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

Theorem 3.1.1 (Existence and Uniqueness for Linear IVPs): Let $P_0(x), P_1(x), \dots, P_{n-1}(x)$ and $g(x)$ be **continuous** on an open interval

I , and let $x_0 \in I$. Then a solution $y(x)$ of the initial-value problem **exists on the entire interval I and is UNIQUE**.

Crucial Engineering Takeaway

: For linear ODEs, discontinuities in the coefficients $P_i(x)$ or $g(x)$ are the **only** points where uniqueness or existence can break down. Unlike nonlinear equations, the interval of definition is completely known beforehand!

B. Boundary-Value Problem (BVP)

Conditions are prescribed at **two or more distinct points** a and b :

$$y(a) = y_0, \quad y(b) = y_1 \quad \text{or} \quad y'(a) = y_0, \quad y(b) = y_1$$

Unlike IVPs, Theorem 3.1.1 does **NOT** apply to BVPs! A boundary-value problem can possess:

1. A **unique solution**,
2. **Infinitely many solutions** (eigenfunction resonance), or
3. **No solution at all!**

Textbook Example

: Solve $y'' + 16y = 0$ with $y(0) = 0$. The general solution is $y(x) = c_1 \cos(4x) + c_2 \sin(4x)$. Condition $y(0) = 0 \implies c_1(1) + c_2(0) = 0 \implies c_1 = 0$, so $y(x) = c_2 \sin(4x)$.

- If boundary condition is $y(\pi/2) = 0$: $c_2 \sin(2\pi) = 0 \implies 0 = 0$. This holds for **any** constant c_2 . Hence, **infinitely many solutions** exist!
- If boundary condition is $y(\pi/2) = 1$: $c_2 \sin(2\pi) = 1 \implies 0 = 1$ (Contradiction!). Hence, **no solution** exists!

2.2 Linear Independence & The Wronskian Determinant

A set of functions $f_1(x), f_2(x), \dots, f_n(x)$ is **linearly dependent** on an interval I if there exist constants $c_1, c_2, \dots, c_n$, not all zero, such that:

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) \equiv 0 \quad \forall x \in I$$

If the only constants that satisfy this identity are $c_1 = c_2 = \dots = c_n = 0$, the functions are **linearly independent**.

The Wronskian Determinant:

For n real-valued functions possessing at least $n - 1$ derivatives on I , the **Wronskian** is the determinant:

$$W(f_1, f_2, \dots, f_n)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_n(x) & f_1'(x) & f_2'(x) & \dots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{vmatrix}$$

For two functions y_1, y_2 :

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 & y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$$

Theorem 3.1.5 (Test for Linearly Independent Solutions): Let $y_1, y_2, \dots, y_n$ be n solutions of the homogeneous n -th order linear differential equation $L(y) = 0$ on interval I . The set of solutions is **linearly independent** on I if and only if:

$$W(y_1, y_2, \dots, y_n)(x) \neq 0 \quad \text{for EVERY } x \in I$$

Abel's Formula for the Wronskian:

If y_1, y_2 are solutions of $y'' + P(x)y' + Q(x)y = 0$, then their Wronskian satisfies the first-order differential equation:

$$\frac{dW}{dx} + P(x)W = 0 \implies W(x) = C e^{-\int P(x) dx}$$

Immediate Consequence

: Either $W(x) = 0$ for all $x \in I$ (if $C = 0$), or $W(x)$ is never zero on I (since the exponential is strictly positive). The Wronskian of solutions to a linear ODE can **never change sign or cross zero** on I !

3. Analytical Methods for Homogeneous Equations (§3.2, §3.3, §3.6)

3.1 Reduction of Order (§3.2)

If one non-zero solution $y_1(x)$ of the second-order homogeneous linear ODE:

$$y'' + P(x)y' + Q(x)y = 0$$

is known, a second linearly independent solution $y_2(x)$ can be discovered by substituting:

$$y_2(x) = u(x)y_1(x)$$

First-Principles Step-by-Step Derivation:

1. **Compute Derivatives of y_2 :**

$$\begin{aligned} y_2' &= u'y_1 + uy_1' \\ y_2'' &= u''y_1 + 2u'y_1' + uy_1'' \end{aligned}$$

2. **Substitute into the ODE:**

$$(u''y_1 + 2u'y_1' + uy_1'') + P(x)(u'y_1 + uy_1') + Q(x)(uy_1) = 0$$

3. **Regroup Coefficients of u'' , u' , u :**

$$y_1u'' + (2y_1' + Py_1)u' + \underbrace{(y_1'' + Py_1' + Qy_1)}_{=0 \text{ because } y_1 \text{ is a solution!}} u = 0$$

The term multiplying u vanishes identically!

$$y_1u'' + (2y_1' + Py_1)u' = 0$$

4. **Reduce to First-Order via $w = u'$:**

$$y_1w' + (2y_1' + Py_1)w = 0 \implies \frac{w'}{w} = -2\frac{y_1'}{y_1} - P(x)$$

5. **Integrate:**

$$\ln |w| = -2 \ln |y_1| - \int P(x)dx = \ln |y_1|^{-2} - \int P(x)dx \implies w = u' = \frac{e^{-\int P(x)dx}}{y_1^2}$$

6. **Integrate to find $u(x)$ and $y_2(x)$:**

$$\begin{aligned} u(x) &= \int \frac{e^{-\int P(x)dx}}{[y_1(x)]^2} dx \\ y_2(x) &= y_1(x) \int \frac{e^{-\int P(x)dx}}{[y_1(x)]^2} dx \end{aligned}$$

3.2 Homogeneous Linear Equations with Constant Coefficients (§3.3)

Consider:

$$ay'' + by' + cy = 0, \quad a \neq 0$$

Since $\frac{d}{dx}[e^{rx}] = re^{rx}$, trial solution $y = e^{rx}$ yields:

$$ar^2e^{rx} + bre^{rx} + ce^{rx} = 0 \implies e^{rx}(ar^2 + br + c) = 0$$

Since $e^{rx} \neq 0$ for all real x , we obtain the **Auxiliary (Characteristic) Equation**:

$$ar^2 + br + c = 0$$

Roots are determined by the quadratic formula: $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Roots of Characteristic Equation: $ar^2 + br + c = 0$

└ Case 1: Discriminant $\Delta = b^2 - 4ac > 0$ (Distinct Real Roots)

└└ $y(x) = c_1 e^{(r_1 x)} + c_2 e^{(r_2 x)}$

└ Case 2: Discriminant $\Delta = b^2 - 4ac = 0$ (Repeated Real Root: $r_1 = r_2 = r$)

└└ $y(x) = (c_1 + c_2 x)e^{(rx)}$

└ Case 3: Discriminant $\Delta = b^2 - 4ac < 0$ (Complex Conjugate Roots: $r = \alpha \pm i\beta$)

└└ $y(x) = e^{(\alpha x)}[c_1 \cos(\beta x) + c_2 \sin(\beta x)]$

Detailed Proof of Case 3 (Euler's Formula Transition):

When $r = \alpha \pm i\beta$ with $\alpha = -\frac{b}{2a}$ and $\beta = \frac{\sqrt{4ac - b^2}}{2a}$:

$$y_{\text{complex}} = C_1 e^{(\alpha + i\beta)x} + C_2 e^{(\alpha - i\beta)x} = e^{\alpha x} [C_1 e^{i\beta x} + C_2 e^{-i\beta x}]$$

Using Euler's identity $e^{\pm i\beta x} = \cos(\beta x) \pm i \sin(\beta x)$:

$$y = e^{\alpha x} [C_1 (\cos \beta x + i \sin \beta x) + C_2 (\cos \beta x - i \sin \beta x)]$$

$$y = e^{\alpha x} [(C_1 + C_2) \cos \beta x + i(C_1 - C_2) \sin \beta x]$$

Defining real constants $c_1 = C_1 + C_2$ and $c_2 = i(C_1 - C_2)$:

$$y(x) = e^{\alpha x} [c_1 \cos(\beta x) + c_2 \sin(\beta x)]$$

3.3 Cauchy-Euler Equations (§3.6)

An equation of the form:

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 x \frac{dy}{dx} + a_0 y = g(x)$$

is called a **Cauchy-Euler (equidimensional) equation**. The degree of each power x^k matches the order of the derivative $\frac{d^k y}{dx^k}$.

The Second-Order Cauchy-Euler Solution Engine:

$$ax^2 \frac{d^2 y}{dx^2} + bx \frac{dy}{dx} + cy = 0, \quad x > 0$$

Substitute trial solution $y = x^m$:

$$y' = mx^{m-1}, \quad y'' = m(m-1)x^{m-2}$$

$$ax^2[m(m-1)x^{m-2}] + bx[mx^{m-1}] + cx^m = 0$$

$$[am(m-1) + bm + c]x^m = 0$$

Since $x^m \neq 0$ for $x > 0$, the **Cauchy-Euler Auxiliary Equation** is:

$$am(m-1) + bm + c = 0 \iff am^2 + (b-a)m + c = 0$$

⚠ CRITICAL EXAM WARNING: Notice the term $(b-a)m$. The first derivative xy' contributes bm , but the second derivative $x^2 y''$ contributes $am(m-1) = am^2 - am$. Missing the $-am$ term is the single most common student error in ENGR 213!

Three Solution Regimes for Cauchy-Euler:

1. **Case 1: Distinct Real Roots** ($m_1 \neq m_2$):

$$y(x) = c_1 x^{m_1} + c_2 x^{m_2}$$

2. **Case 2: Repeated Real Root** ($m_1 = m_2 = m$):

$$y(x) = c_1 x^m + c_2 x^m \ln x = (c_1 + c_2 \ln x) x^m$$

3. **Case 3: Complex Conjugate Roots** ($m = \alpha \pm i\beta$): Recall that $x^{i\beta} = (e^{\ln x})^{i\beta} = e^{i\beta \ln x} = \cos(\beta \ln x) + i \sin(\beta \ln x)$.

$$y(x) = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$$

4. Non-Homogeneous Differential Equations (§3.4, §3.5)

The general solution of a non-homogeneous linear ODE $L(y) = g(x)$ is:

$$y(x) = y_c(x) + y_p(x)$$

where $y_c(x)$ is the complementary solution of $L(y) = 0$, and $y_p(x)$ is any particular solution of $L(y) = g(x)$.

4.1 Method of Undetermined Coefficients (§3.4)

Applicable when the differential equation has **constant coefficients** and the forcing function $g(x)$ is a linear combination of:

- Polynomials (x^k)
- Exponentials ($e^{\alpha x}$)
- Sines and Cosines ($\sin \beta x, \cos \beta x$)
- Products of the above ($x^k e^{\alpha x} \cos \beta x$)

Forcing Term $g(x)$	Trial Particular Solution $y_p(x)$
1 (constant)	A
$5x + 7$	$Ax + B$
$3x^2 - 2$	$Ax^2 + Bx + C$
e^{5x}	Ae^{5x}
$\sin(3x)$ or $\cos(3x)$	$A \cos(3x) + B \sin(3x)$
xe^{2x}	$(Ax + B)e^{2x}$
$e^{3x} \cos(2x)$	$e^{3x}(A \cos 2x + B \sin 2x)$
$x^2 \sin(4x)$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$

The Duplication Rule (Rule of Multiplication):

If any term in the trial particular solution y_p duplicates a term in the complementary solution $y_c(x)$, that entire trial family must be multiplied by x^s , where s is the **smallest positive integer** ($s = 1$ or $s = 2$) that eliminates all duplication with $y_c(x)$.

4.2 Method of Variation of Parameters (§3.5)

Variation of parameters is the universal, powerful method that solves $y'' + P(x)y' + Q(x)y = f(x)$ regardless of whether $f(x)$ is amenable to undetermined coefficients (e.g., $f(x) = \tan x, \sec x, \frac{1}{x}, \ln x, \frac{e^x}{x}$).

First-Principles Step-by-Step Derivation:

We replace the constants in $y_c = c_1 y_1 + c_2 y_2$ with unknown variable parameters $u_1(x), u_2(x)$:

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

1. Differentiate:

$$y'_p = (u'_1 y_1 + u'_2 y_2) + (u_1 y'_1 + u_2 y'_2)$$

To simplify and avoid second derivatives of u_1, u_2 , impose the first constraint:

$$u'_1 y_1 + u'_2 y_2 = 0$$

Then $y'_p = u_1 y'_1 + u_2 y'_2$.

2. Differentiate again:

$$y''_p = u'_1 y'_1 + u_1 y''_1 + u'_2 y'_2 + u_2 y''_2$$

3. Substitute y_p, y'_p, y''_p into $y'' + Py' + Qy = f(x)$:

$$(u'_1 y'_1 + u_1 y''_1 + u'_2 y'_2 + u_2 y''_2) + P(u_1 y'_1 + u_2 y'_2) + Q(u_1 y_1 + u_2 y_2) = f(x)$$

Regrouping: $\underbrace{u_1(y''_1 + P y'_1 + Q y_1)}_{=0} + \underbrace{u_2(y''_2 + P y'_2 + Q y_2)}_{=0} + (u'_1 y'_1 + u'_2 y'_2) = f(x)$

$$\underbrace{u_1(y''_1 + P y'_1 + Q y_1)}_{=0} + \underbrace{u_2(y''_2 + P y'_2 + Q y_2)}_{=0} + (u'_1 y'_1 + u'_2 y'_2) = f(x)$$

$$(u'_1 y'_1 + u'_2 y'_2) = f(x)$$

Yielding the second constraint :

$$u'_1 y'_1 + u'_2 y'_2 = f(x)$$

4. Solve the linear system for u'_1 and u'_2 via Cramer's Rule:

$$\begin{cases} y_1 u'_1 + y_2 u'_2 = 0 \\ y'_1 u'_1 + y'_2 u'_2 = f(x) \end{cases} \implies \begin{bmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{bmatrix} \begin{bmatrix} u'_1 \\ u'_2 \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}$$

$$\mathbf{u}'_1(\mathbf{x}) = -\frac{\mathbf{y}_2(\mathbf{x})\mathbf{f}(\mathbf{x})}{\mathbf{W}(\mathbf{y}_1, \mathbf{y}_2)}, \quad \mathbf{u}'_2(\mathbf{x}) = \frac{\mathbf{y}_1(\mathbf{x})\mathbf{f}(\mathbf{x})}{\mathbf{W}(\mathbf{y}_1, \mathbf{y}_2)}$$

Integrate to find $u_1(x)$ and $u_2(x)$, and assemble:

$$y_p(x) = y_1(x) \int -\frac{y_2(x)f(x)}{W(x)} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(x)} dx$$

5. Applied Mechanical Oscillations & Resonance (§3.8)

5.1 Mass-Spring-Damper Physical Architecture

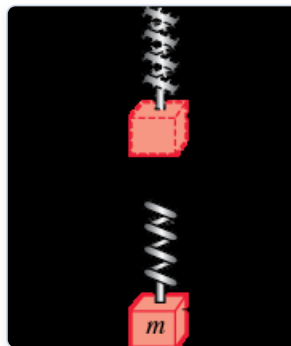


Figure 3.1: Mass-spring-damper mechanical architecture showing reference position, equilibrium elongation s , and dynamic displacement $x(t)$ — from Textbook 7th Ed. (Fig. 3.8.2).

Derivation of the Governing Equation from Newton's Second Law:

- Let a mass m suspend vertically from a spring with spring constant k (Hooke's Law).
- At static equilibrium, the downward gravitational force balances the upward spring tension:

$$mg = ks$$

- Displacing the mass downward by $x(t)$:
 - Net spring restoring force: $F_{\text{spring}} = -k(s + x) = -ks - kx$
 - Viscous damping force (dashpot): $F_{\text{damping}} = -\beta \frac{dx}{dt}$ ($\beta > 0$)
 - External driving force: $F_{\text{ext}} = f(t)$
- Applying Newton's Second Law $\sum F = ma = m \frac{d^2x}{dt^2}$:

$$m \frac{d^2x}{dt^2} = mg - k(s + x) - \beta \frac{dx}{dt} + f(t)$$

Since $mg - ks = 0$:

$$m \frac{d^2x}{dt^2} + \beta \frac{dx}{dt} + kx = f(t)$$

Dividing by m and defining $2\lambda = \frac{\beta}{m}$ and $\omega^2 = \frac{k}{m}$:

$$\frac{d^2x}{dt^2} + 2\lambda \frac{dx}{dt} + \omega^2 x = F(t)$$

5.2 Free Damped Vibrations & Regime Classification

Characteristic equation: $r^2 + 2\lambda r + \omega^2 = 0 \implies r = -\lambda \pm \sqrt{\lambda^2 - \omega^2}$.

$$x(t) = \frac{1}{6} \cos(8t - (-0.245)) \quad \text{or} \quad x(t) = \frac{1}{6} \cos(8t + 0.245). \quad (9')$$

Graphical Interpretation

FIGURE 3.8.4(a) illustrates the mass in Example 2 going through approximately two complete cycles of motion. Reading left to right, the first five positions marked with black dots in the figure correspond, respectively,

- to the initial position ($x = \frac{2}{3}$) of the mass below the equilibrium position,
- to the mass passing through the equilibrium position ($x = 0$) for the first time heading upward,

Figure 3.2: Comparison of displacement trajectories for Overdamped, Critically Damped, and Underdamped free motion — from Textbook 7th Ed. (Fig. 3.8.4).

1. Overdamped Motion ($\lambda^2 - \omega^2 > 0 \iff \beta^2 > 4mk$)

- Damping dominates elasticity (heavy oil immersion).
- Real distinct negative roots $r_1, r_2 < 0$:

$$x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

- Smooth exponential return to equilibrium with at most one zero-crossing (no oscillation).

2. Critically Damped Motion ($\lambda^2 - \omega^2 = 0 \iff \beta^2 = 4mk$)

- Perfect balance; smallest damping that completely suppresses oscillation.
- Repeated root $r = -\lambda$:

$$x(t) = (c_1 + c_2 t)e^{-\lambda t}$$

- Returns to equilibrium faster than any overdamped system without oscillating (ideal for automotive shock absorbers and galvanometer dials).

3. Underdamped Motion ($\lambda^2 - \omega^2 < 0 \iff \beta^2 < 4mk$)

- Elasticity dominates damping (spring oscillations in air).
- Complex roots $r = -\lambda \pm i\omega_d$, where $\omega_d = \sqrt{\omega^2 - \lambda^2}$ is the **quasi-frequency**:

$$x(t) = e^{-\lambda t} [c_1 \cos(\omega_d t) + c_2 \sin(\omega_d t)] = Ae^{-\lambda t} \sin(\omega_d t + \phi)$$

- True oscillatory motion bounded inside an exponentially decaying envelope curve $\pm Ae^{-\lambda t}$.
- Quasi-period: $T = \frac{2\pi}{\omega_d} = \frac{2\pi}{\sqrt{\omega^2 - \lambda^2}}$.

5.3 Forced Oscillations & Catastrophic Pure Resonance

Consider an undamped oscillator ($\beta = 0$) driven by an external periodic sinusoidal force at driving frequency γ :

$$\frac{d^2 x}{dt^2} + \omega^2 x = F_0 \cos(\gamma t), \quad F_0 = \frac{f_0}{m}$$

1. **Non-Resonant Case** ($\gamma \neq \omega$): Complementary solution: $x_c = c_1 \cos \omega t + c_2 \sin \omega t$. Trial particular solution: $x_p = A \cos \gamma t + B \sin \gamma t \implies x_p(t) = \frac{F_0}{\omega^2 - \gamma^2} \cos(\gamma t)$. General solution:

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t + \frac{F_0}{\omega^2 - \gamma^2} \cos(\gamma t)$$

Motion is bounded. When $\gamma \approx \omega$, interference between the two close frequencies produces **beats**.

2. **Pure Resonance Case** ($\gamma = \omega$): The driving frequency matches the natural frequency of the structure! Trial form $A \cos \omega t + B \sin \omega t$ collides with x_c . Applying the **Duplication Rule**, multiply by t :

$$x_p(t) = t(A \cos \omega t + B \sin \omega t)$$

Substituting into $x'' + \omega^2 x = F_0 \cos \omega t$:

$$x'_p = (A \cos \omega t + B \sin \omega t) + t(-\omega A \sin \omega t + \omega B \cos \omega t)$$

$$x''_p = 2(-\omega A \sin \omega t + \omega B \cos \omega t) + t(-\omega^2 A \cos \omega t - \omega^2 B \sin \omega t)$$

$$x''_p + \omega^2 x_p = -2\omega A \sin \omega t + 2\omega B \cos \omega t = F_0 \cos \omega t$$

Equating coefficients:

$$-2\omega A = 0 \implies A = 0$$

$$2\omega B = F_0 \implies B = \frac{F_0}{2\omega}$$

$$\mathbf{x_p(t) = \frac{F_0}{2\omega} t \sin(\omega t)}$$

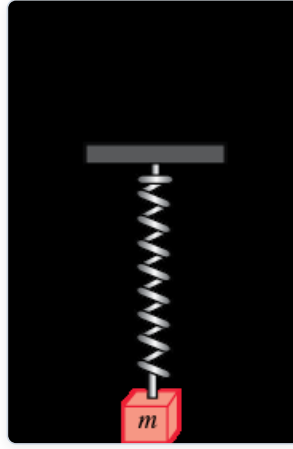


Figure 3.3: Pure resonance response curve showing linear amplitude envelope $\pm \frac{F_0}{2\omega} t$ growing unbounded as $t \rightarrow \infty$ — from Textbook 7th Ed. (Fig. 3.8.13).

Engineering Physical Consequences:

The amplitude $\frac{F_0}{2\omega} t$ grows **linearly with time without bound!** In physical engineering, this results in mechanical destruction (e.g., the Tacoma Narrows bridge collapse, aircraft flutter, turbine blade failure).

6. Fully Worked Master Archetypes

Archetype 6.1: Variation of Parameters with Tangent Forcing

Problem: Find the general solution of $y'' + 9y = \tan(3x)$ on $(-\frac{\pi}{6}, \frac{\pi}{6})$.

Step-by-Step Solution:

- **Step 1: Solve the Homogeneous Equation:**

$$r^2 + 9 = 0 \implies r = \pm 3i \implies y_1 = \cos(3x), \quad y_2 = \sin(3x)$$

- **Step 2: Calculate the Wronskian:**

$$W = \begin{vmatrix} \cos(3x) & \sin(3x) \\ -3\sin(3x) & 3\cos(3x) \end{vmatrix} = 3\cos^2(3x) - (-3\sin^2(3x)) = 3(\cos^2 3x + \sin^2 3x) = 3$$

- **Step 3: Calculate $u'_1(x)$ and Integrate:**

$$u'_1(x) = -\frac{y_2 f(x)}{W} = -\frac{\sin(3x) \tan(3x)}{3} = -\frac{1}{3} \frac{\sin^2(3x)}{\cos(3x)} = -\frac{1}{3} \frac{1 - \cos^2(3x)}{\cos(3x)} = -\frac{1}{3} [\sec(3x) - \cos(3x)]$$

$$u_1(x) = -\frac{1}{3} \int \sec(3x) dx + \frac{1}{3} \int \cos(3x) dx = -\frac{1}{9} \ln |\sec(3x) + \tan(3x)| + \frac{1}{9} \sin(3x)$$

- **Step 4: Calculate $u'_2(x)$ and Integrate:**

$$u'_2(x) = \frac{y_1 f(x)}{W} = \frac{\cos(3x) \tan(3x)}{3} = \frac{\sin(3x)}{3}$$

$$u_2(x) = \frac{1}{3} \int \sin(3x) dx = -\frac{1}{9} \cos(3x)$$

- **Step 5: Assemble $y_p = u_1 y_1 + u_2 y_2$:**

$$y_p(x) = \left[-\frac{1}{9} \ln |\sec 3x + \tan 3x| + \frac{1}{9} \sin 3x \right] \cos 3x + \left[-\frac{1}{9} \cos 3x \right] \sin 3x$$

$$y_p(x) = -\frac{1}{9} \cos(3x) \ln |\sec 3x + \tan 3x| + \underbrace{\frac{1}{9} \sin 3x \cos 3x - \frac{1}{9} \cos 3x \sin 3x}_{=0}$$

$$y_p(x) = -\frac{1}{9} \cos(3x) \ln |\sec(3x) + \tan(3x)|$$

- **Final General Solution:**

$$y(x) = c_1 \cos(3x) + c_2 \sin(3x) - \frac{1}{9} \cos(3x) \ln |\sec(3x) + \tan(3x)|$$

Archetype 6.2: Cauchy-Euler Inhomogeneous Equation IVP

Problem: Solve $x^2 y'' - 4xy' + 6y = 3x^{-3}$ with $y(1) = 0$; $y'(1) = 1$ for $x > 0$.

Step-by-Step Solution:

- **Step 1: Solve the Homogeneous Cauchy-Euler Equation:**

$$m(m-1) - 4m + 6 = 0 \implies m^2 - 5m + 6 = 0 \implies (m-2)(m-3) = 0$$

$$m_1 = 2, \quad m_2 = 3 \implies y_c = c_1 x^2 + c_2 x^3$$

- **Step 2: Find Particular Solution $y_p(x)$:** Since $g(x) = 3x^{-3}$ is a power function, test $y_p = Ax^{-3}$:

$$y_p' = -3Ax^{-4}, \quad y_p'' = 12Ax^{-5}$$

Substitute into the LHS:

$$x^2(12Ax^{-5}) - 4x(-3Ax^{-4}) + 6(Ax^{-3}) = (12 + 12 + 6)Ax^{-3} = 30Ax^{-3}$$

Equate to the RHS:

$$30Ax^{-3} = 3x^{-3} \implies 30A = 3 \implies A = \frac{1}{10}$$

$$y_p(x) = \frac{1}{10}x^{-3}$$

- **Step 3: Assemble General Solution:**

$$y(x) = c_1 x^2 + c_2 x^3 + \frac{1}{10}x^{-3}$$

- **Step 4: Compute Derivative and Apply Initial Conditions:**

$$y'(x) = 2c_1 x + 3c_2 x^2 - \frac{3}{10}x^{-4}$$

$$\circ y(1) = 0 \implies c_1 + c_2 + \frac{1}{10} = 0 \implies c_1 + c_2 = -\frac{1}{10}$$

$$\circ y'(1) = 1 \implies 2c_1 + 3c_2 - \frac{3}{10} = 1 \implies 2c_1 + 3c_2 = \frac{13}{10}$$

- **Step 5: Solve the 2×2 Linear System:** Multiply the first equation by 2: $2c_1 + 2c_2 = -\frac{2}{10}$. Subtract from the second equation:

$$(2c_1 + 3c_2) - (2c_1 + 2c_2) = \frac{13}{10} - \left(-\frac{2}{10}\right) \implies c_2 = \frac{15}{10} = \frac{3}{2}$$

$$c_1 = -\frac{1}{10} - c_2 = -\frac{1}{10} - \frac{3}{2} = -\frac{16}{10} = -\frac{8}{5}$$

- **Final Explicit Solution:**

$$y(x) = -\frac{8}{5}x^2 + \frac{3}{2}x^3 + \frac{1}{10x^3}$$

7. Diagnostic Traps & Common Exam Pitfalls

- **⚠ Trap 1: The Cauchy-Euler $m^2 - m$ Subtraction:** When substituting $y = x^m$ into $ax^2y'' + bxy' + cy = 0$, students frequently write $am^2 + bm + c = 0$.

Correction

: Because $\frac{d^2}{dx^2}[x^m] = m(m-1)x^{m-2}$, the auxiliary equation is $am(m-1) + bm + c = am^2 + (b-a)m + c = 0$!

- **⚠ Trap 2: Forgetting to Put the ODE in Standard Form for Variation of Parameters:** If given $2y'' + 4y' + 8y = \sec x$, the forcing function is **NOT** $\sec x$. You **MUST** divide by the leading coefficient 2 first:

$$y'' + 2y' + 4y = \frac{1}{2} \sec x \implies f(x) = \frac{1}{2} \sec x$$

Forgetting to divide by 2 doubles y_p and ruins the entire calculation.

- **⚠ Trap 3: Undetermined Coefficients Duplication with Repeated Roots:** If $y_c = (c_1 + c_2x)e^{2x}$ and the driving force is $g(x) = (x - 2)e^{2x}$, multiplying by x gives $(Ax^2 + Bx)e^{2x}$, which **still** collides with c_2xe^{2x} ! You must multiply by x^2 :

$$y_p = (Ax^3 + Bx^2)e^{2x}$$